C. Parquet

time limit per test: 2 seconds
memory limit per test: 256 megabytes

Once Bob decided to lay a parquet floor in his living room. The living room is of size $n \times m$ metres. Bob had planks of three types: a planks 1×2 meters, b planks 2×1 meters, and c planks 2×2 meters. Help Bob find out, if it is possible to parquet the living room with such a set of planks, and if it is possible, find one of the possible ways to do so. Bob doesn't have to use all the planks.

Input

The first input line contains 5 space-separated integer numbers n, m, a, b, c $(1 \le n, m \le 100, 0 \le a, b, c \le 10^4)$, n and m — the living room dimensions, a, b and c — amount of planks 1×2, 2×1 и 2×2 respectively. It's not allowed to turn the planks.

Output

If it is not possible to parquet the room with such a set of planks, output `IMPOSSIBLE`. Otherwise output one of the possible ways to parquet the room — output n lines with m lower-case Latin letters each. Two squares with common sides should contain the same letters, if they belong to one and the same plank, and different letters otherwise. Different planks can be marked with one and the same letter (see examples). If the answer is not unique, output any.

Examples

input Copy

```
2 6 2 2 1
```

output Copy

```
aabcca
aabdda
```

input Copy

```
1 1 100 100 100
```

output Copy

```
IMPOSSIBLE
```

input Copy

```
4 4 10 10 10
```

output Copy

```
aabb
aabb
bbaa
bbaa
```

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 26 (Codeforces format)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
372300844	Apr/24/2026 05:22	Accepted
372300706	Apr/24/2026 05:18	Accepted
372300663	Apr/24/2026 05:17	Accepted

372300563	Apr/24/2026 05:14	Wrong answer on test 1

→ Problem tags

combinatorics | constructive algorithms | greedy | implementation | *2000

No tag edit access

→ Contest materials

- Announcement
- Tutorial #1
- Tutorial #2 (ru)

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